CSE 2025 ll Rancho Bernardo High School

Water Quality

Name(s)_____

Score: __/40

Section 1: Freshwater ecology

1. What is a keystone species? (1 Pt)
    a. A species that has the largest impact on an ecosystem
    b. A species that is not native from that ecosystem
    c. A species that is the most common/abundant in the ecosystem
    d. A species that is at the top of the food chain in its ecosystem

2. What are two factors that affect the percent of dissolved oxygen there is inside of a body of water? (2 Pts)

3. What is the primary nutrient that causes eutrophication? (1 Pt)
    a. calcium
    b. Iron
    c. Nitrogen
    d. Oxygen

4. What is a man made source of turbidity in marine ecosystems? (1 Pt)

5. Which of the following is an effect of an algal bloom? (1 Pt)
    a. An increased amount of food for organisms in the ecosystem
    b. An increase in biodiversity
    c. No change
    d. A decrease in biodiversity due to the creation of dead zones

6. What pH level is considered normal for the rivers in the United States? (1 Pt)
    a. 5.5-7.5
    b. 6.5-8.5
    c. 7.5-9.5
    d. 4.5-6.5

7. Which one of these factors can be affected by temperature in the water? (1 Pt)
    a. Photosynthesis of aquatic plants
    b. Color of the water
    c. Pressure
    d. Nitrogen levels

8. Name one disease that is associated with high levels of fecal coliform in drinking water. (1 Pt)

9. What is a natural source of phosphates for marine ecosystems? (1 Pt)
    a. Erosion of rocks and minerals
    b. Fertilizers
    c. Mine runoffs
    d. Industrial and cleaning compounds

10. What does “BOD” stand for, and what does it measure? (1 Pt)

11. What are 3 effects of turbidity? (3 Pts)

Section 2: Macroflora and Fauna Identification

The next 5 questions are related to the organism depicted below:



1. Identify the organism depicted above (1 pt).
2. What order is the organism depicted above associated with? (1 pt)
3. What method does the organism use in order to deliver its offspring? (1 pt)
4. What term is used to describe a dormancy in egg hatching in order to await for more suitable conditions? (1 pt).
5. What is the process used to progress from different instars by shedding their exoskeleton called? (1 pt)

The next 4 questions are related to the organism depicted below.



6. Identify the organism depicted above (1 pt).
7. Describe the typical environment that a cranefly could be found in. (1 pt)
8. What does the organism above indicate about water quality? (1 pt)
9. What is the typical lifespan of an adult of this organism? (1 pt)

The next 4 questions are related to the organism depicted below.



10. Identify the organism depicted above (1 pt).
11. What is the primary feeding strategy of the organism? (1 pt).
12. How do flying boatmen breathe underwater? (1 pt)
13. How do flying boatmen attract mates? (1 pt)

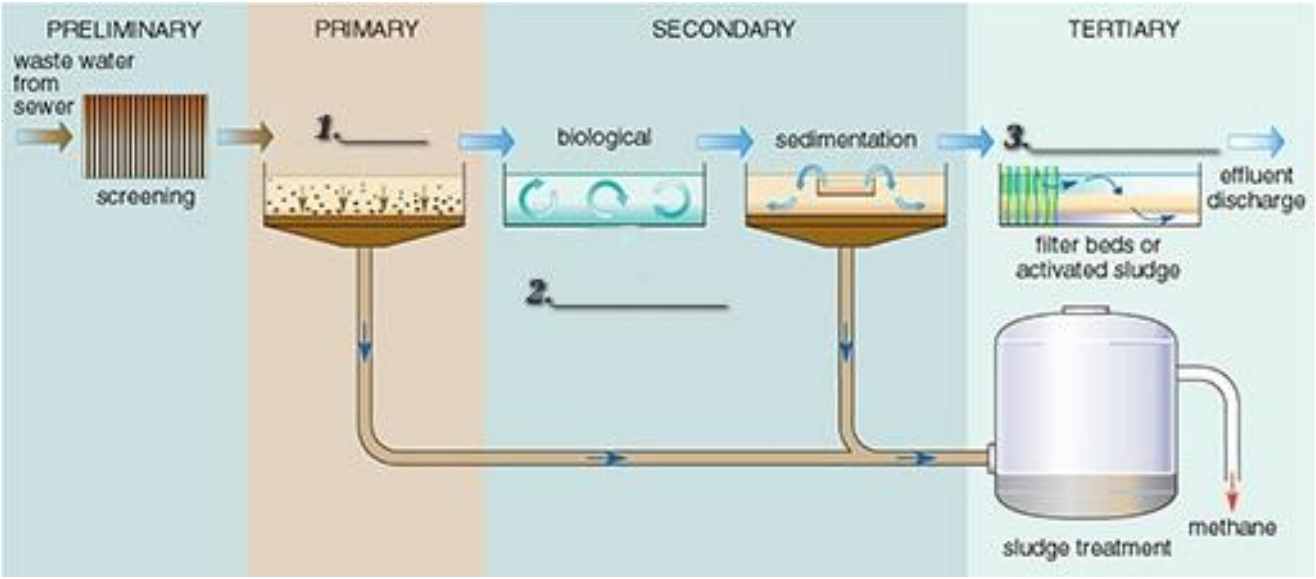
The next 4 questions are related to the organism depicted below.



14. Identify the organism depicted above (1 pt).
15. How does it primarily reproduce? (1 pt)
16. What species of nitrogen fixing bacteria are located at the base of the petioles? (2 pt)
17. How does this organism affect water quality? (1 pt)

Section 3: Analysis

1-3. Name the process that is blanked out (1 Pt for each blank)



4. How can warm thermal pollution affect the population of aquatic plants that live in that particular environment? (1 Pt)

5. Despite often being a negative impact, warm thermal pollution can affect populations of certain species in positive ways. Name one species that benefits from warm thermal pollution. (1 Pt)

6. The H+ concentration inside of a container of water is 1.0x10^-5.8. What is the pH of the water and can the water support life? (2 Pts)

7. If the salinity of water is 1.2 ppt, the water is _____. (1 Pt)
   a. Fresh water
   b. Brackish water
   c. Saline Water
   d. Brine

8. What level of TDS is considered freshwater? (1 Pt)
   a. 1000-30000 ppm
   b. 30000-40000 ppm
   c. <1000 ppm
   d. >40000 ppm